Class 12 SQL cheat sheet

Page 1 · Define, change and query data

BASICS

`SHOW DATABASES;`	list databases
`SHOW TABLES;`	list tables
`USE db_name;`	open a database
`DESC table_name;`	table structure

DDL · DEFINE STRUCTURE

`CREATE DATABASE db;`	
`CREATE TABLE t(f1 type1, f2 type2);`	
`DROP TABLE t;`	delete table
`TRUNCATE TABLE t;`	delete all rows
`ALTER TABLE t ADD col type;`	add column
`ALTER TABLE t MODIFY col type;`	change type
`ALTER TABLE t DROP col;`	remove column
`ALTER TABLE t CHANGE old new type;`	rename column
`ALTER TABLE t RENAME TO t2;`	rename table

DML · WORK WITH DATA

`INSERT INTO t(c1, c2) VALUES(v1, v2);`	
`UPDATE t SET f = value WHERE cond;`	
`DELETE FROM t WHERE cond;`	
`SELECT f1, f2 FROM t;`	SELECT * = all columns

CLAUSE ORDER

`SELECT … FROM … WHERE … GROUP BY …`	
`HAVING … ORDER BY … LIMIT …;`	always in this order

FILTERS

`WHERE f BETWEEN a AND b`	
`WHERE f LIKE 'A%'`	starts with A
`WHERE f IN (v1, v2)`	
`WHERE f IS NULL / IS NOT NULL`	
`SELECT DISTINCT f FROM t;`	no duplicates
`ORDER BY f DESC, g ASC`	sorting

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pavithrasajunair-ship-it.github.io/sql-web

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Page 2 · Aggregates, joins, functions, Python + MySQL

AGGREGATE FUNCTIONS

```
SUM(f) AVG(f) MIN(f) MAX(f)
COUNT(*) · COUNT(f)                          COUNT(f) skips NULLs
SELECT f, COUNT(*) FROM t GROUP BY f HAVING cond;
```

JOINS

```
SELECT * FROM t1, t2;                        Cartesian product
… FROM t1, t2 WHERE t1.c = t2.c;             equi-join
… FROM t1 JOIN t2 ON t1.c = t2.c;
… FROM t1 NATURAL JOIN t2;                   same-named columns
```

CONSTRAINTS

```
PRIMARY KEY                                  unique, never NULL
UNIQUE · NOT NULL · DEFAULT 'x'
CHECK(age >= 18) · AUTO_INCREMENT
FOREIGN KEY (c) REFERENCES t2(id)            links tables
```

FUNCTIONS

```
UPPER() LOWER() LENGTH()
LEFT(s,n) RIGHT(s,n) MID(s,pos,n)
INSTR(s,sub) TRIM(s)
POWER(a,b) ROUND(x,d) MOD(a,b)
NOW() CURDATE() DAY() MONTH() YEAR()
DAYNAME() MONTHNAME()
```

RELATIONAL MODEL

```
Degree = columns · Cardinality = rows
Tuple = row · Attribute = column
Candidate key → Primary key (chosen) + Alternate keys
Foreign key refers to another table's primary key
```

PYTHON + MYSQL

```
import mysql.connector as sql
con = sql.connect(host=…, user=…, passwd=…, database=…)
cur = con.cursor(); cur.execute(query)
fetchone() fetchall() fetchmany(n)           rowcount
con.commit(); con.close()
```